



Xiaomi's personal SoC?

Xiaomi might join Samsung, Apple and Huawei in the elite club of smartphone makers building their own processors. <http://dgit.in/xiaomiSoC>

Projector toy for kids

FollowGrams is a smart projector toy for kids lets them trace any smartphone photo and project it on paper. <http://dgit.in/projectKid>

competitors. Of course, the Z11 Mini has its weaknesses in UI, speed and battery life, but if camera is your main requirement, we think you should consider this one.

The Honor 6X and Coolpad Cool 1 are our second and third choices in the camera department. Between these two, the Cool 1 loses white balance a bit too often, which lands it in the second place. The Honor 6X produces nicely saturated photos with its dual-camera, although both are unable to produce the details that the Nubia Z11 Mini can produce. Where the Honor and Coolpad devices excel though, is in producing bokeh effects, by phone standards anyway. Don't go expecting DSLR quality bokeh from these.

Battery

With performance and camera dealt with, our comparison also takes into account the battery life of these devices. For this, we used the PC Mark Work 2.0 battery benchmark to get side-by-side results of each device. The PC Mark test emulates real world usage, by playing videos, browsing and editing documents on the device. It noted the time taken for the battery to drop from 80% to 20%, which is the sweet spot for battery performance. We have obviously tested the real world battery performance of all these devices when reviewing them, so we have a direct parallel for each PC Mark score that we got.

The battery test is another arena where Lenovo steals the show. The Chinese manufacturer's Lenovo P2 came out far ahead of anyone else in this department, lasting almost 21 hours on the PC Mark test. Second, and closest to this device was the

Asus Zenfone 3s Max, which came in at just over 12 hours, while the ZTE Blade A2 Plus followed, lasting 11.8 hours on the PC Mark test.

In terms of real world, you can expect a full day's (not a work day) worth of battery life out of the Zenfone 3s Max and the ZTE Blade A2 Plus, but the Lenovo P2 is perhaps the only smartphone that can carry on



Lenovo Z2 Plus

for more than that. We have been able to make it last for two days, without much effort.

The Lenovo P2 is also better than the other two in other aspects, like speed and camera performance. The Asus and ZTE smartphones make major compromises to churn out battery life, but can't match up. For a battery centric buyer, the Lenovo P2 makes a lot of sense, but you're nearing the 20k mark, since the P2 costs ₹17,999 for the 4GB variant.

Here, we must mention that while these three smart-

phones were battery centric, the 10-20k segment provides decent battery life in general. Phones like the Xiaomi Redmi Note 4, Coolpad Cool 1, Honor 6X, and Xiaomi Mi Max all last for almost a day. With the exception of the Lenovo Phab 2 Plus and Lenovo Z2 Plus, not many produce battery life that we haven't already seen in this segment.

FEATURES

In terms of features, most of these phones are similarly matched. They all have fingerprint sensors and are driven by similarly matched SoCs. However, Huawei's Kirin 655 SoC is noticeably slower than the Snapdragon 650 and 652 powering its primary competitors. As mentioned earlier, the Snapdragon 820 is much faster than all the other SoCs here.

ZTE's choice of the MediaTek Helio P10 for the Nubia N1 is unfortunate, but the company has actually done quite well in optimising performance. The Helio P10 performs much better here than what we've seen on many other smartphones, barring perhaps the Oppo F1 Plus, which is also much more expensive.

The Cortex A72 cores in the Snapdragon 650 and 652 give them a huge boost in terms of raw power, while the Kryo CPU on the 820 is Qualcomm's own formula for success. As we mentioned earlier, Lenovo hasn't taken complete advantage of this, but has been able to churn out more performance than competing devices in this comparison.

We noticed that the two phones running on MediaTek's MT6750 SoCs were the weakest of the lot, showing the most lags and stutters, while the MediaTek MT8783 on the

Lenovo Phab 2 Plus is equally slow. Overall, these three made for the bottom rung in terms of pure performance. The Snapdragon 617 on the Moto G4 Plus is older, but incredibly stable, and Moto has the right background enhancements to make its performance smoother than expected.

In terms of camera resolution, you'll mostly see 13MP sensors on these smartphones. The Xiaomi Mi Max, Moto G4 Plus and the LeEco Le 2 use 16MP sensors, but they aren't evenly matched. The G4 Plus delivers much better image quality than the other two, producing better colours and sharpness. Low light performance is also significantly better on the Moto G4 Plus.

The Coolpad Cool 1 and Honor 6X stand out with their dual-camera sensors. While Coolpad uses two 13MP sensors on the back, the Honor phone uses a 12MP and 2MP camera on the back. Here, though, Coolpad uses its two sensors in RGB and Black and White mode, which should ideally produce more details. However, after thorough testing, we found the Honor 6X to be slightly ahead, even though its 2MP sensor is used only for depth perception and contrast.

Most of these phones are pretty light as well, with the exception of the huge Xiaomi Mi Max and Lenovo Phab 2 Plus, which do not aim to impress in terms of ergonomics.

DISPLAY

The first thing you see on any smartphone is its display. It may have a fast chipset, oodles of RAM and the best camera ever, but it's the display that sets the first impression, and